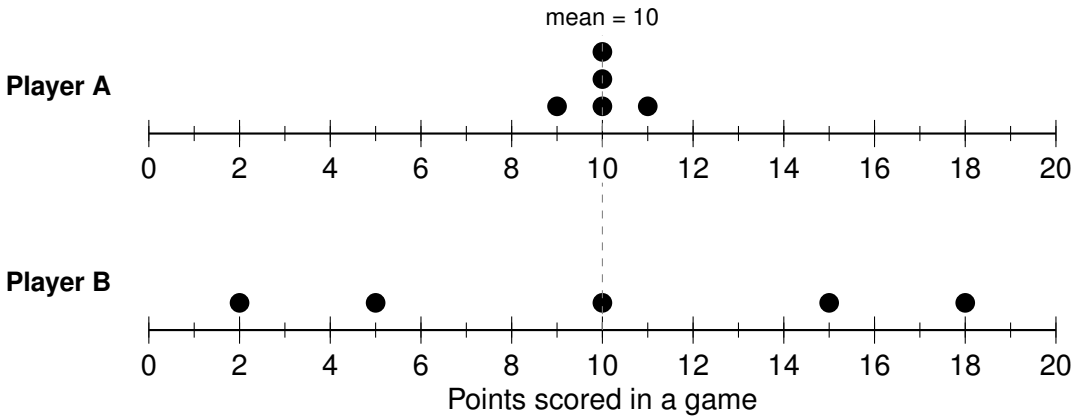
# Standard Deviation KEY

Name _____

Two basketball players both score **10 points** per game on average.




Player A's points are close to the mean. Player B's points are spread far from the mean. The **standard deviation** (SD) measures this.

| |
|---|
| **Standard deviation** is about how far the values usually are from the mean.<br>**Small SD:** values are close to the mean. **Large SD:** values are spread out. |

**How to find the standard deviation** **Example:** 7, 7, 10, 13, 13

1. Find the mean, $\bar{x}$. $\bar{x} = 50 \div 5 = 10$
2. Subtract the mean from each value.
3. Square each result.
4. Add the squares. $\Sigma(x - \bar{x})^2 = 36$
5. Divide by $n - 1$. $36 \div 4 = 9$
   (This number is called the **variance**.)
6. Take the square root. $s = \sqrt{9} = \mathbf{3}$

| $x$ | $x - \bar{x}$ | $(x - \bar{x})^2$ |
|---|---|---|
| 7 | $-3$ | 9 |
| 7 | $-3$ | 9 |
| 10 | 0 | 0 |
| 13 | 3 | 9 |
| 13 | 3 | 9 |
| $\Sigma(x - \bar{x})^2$ | | 36 |

**The formula:** $s = \sqrt{\dfrac{\Sigma(x - \bar{x})^2}{n - 1}}$ ($s$ = the standard deviation, $n$ = how many values)

$\Sigma$ (capital **sigma**) means **add up**. $\Sigma(x - \bar{x})^2$ means: find $(x - \bar{x})^2$ for every value, then add them all.

**Why $n - 1$?** Our data is usually a **sample**, a small part of a bigger group. A sample is usually a little less spread out than the whole group. Dividing by $n - 1$ makes $s$ a little bigger to fix this.

**Remember:**

- The SD is never negative.
- The SD is 0 only when every value is the same.
- The SD has the same units as the data (points, minutes, inches).

Work with a partner. Compare answers when you finish each problem.

1. Find the standard deviation of 4, 6, 8, 10, 12.

| $x$ | $x - \bar{x}$ | $(x - \bar{x})^2$ |
|---|---|---|
| 4 | $-4$ | 16 |
| 6 | $-2$ | 4 |
| 8 | 0 | 0 |
| 10 | 2 | 4 |
| 12 | 4 | 16 |
| | $\Sigma(x - \bar{x})^2$ | 40 |

Mean: $40 \div 5 = 8$

$s = \sqrt{40 \div 4} = \sqrt{10} \approx 3.16$

2. Four quiz scores: 7, 3, 7, 7. Find the standard deviation.

| $x$ | $x - \bar{x}$ | $(x - \bar{x})^2$ |
|---|---|---|
| 7 | 1 | 1 |
| 3 | $-3$ | 9 |
| 7 | 1 | 1 |
| 7 | 1 | 1 |
| | $\Sigma(x - \bar{x})^2$ | 12 |

Mean: $24 \div 4 = 6$

$s = \sqrt{12 \div 3} = \sqrt{4} = 2$

3. Look at the two players on page 1.

   Player A: 9, 10, 10, 11, 10
   Player B: 2, 18, 5, 15, 10

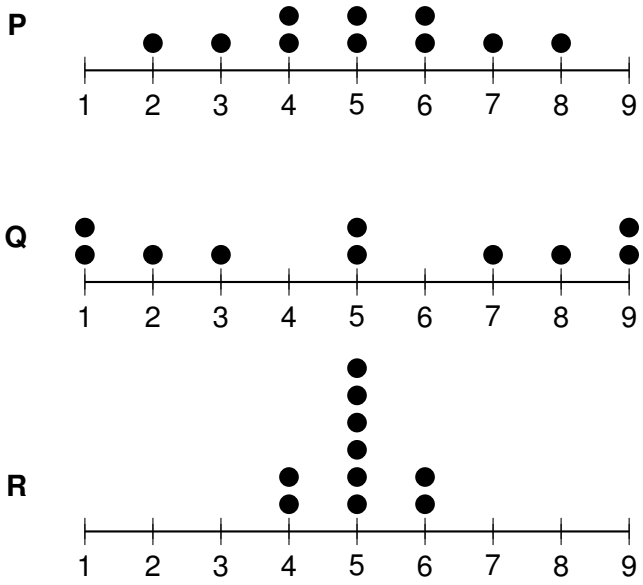
   (a) Find the mean of each player. A: $50 \div 5 = 10$. B: $50 \div 5 = 10$.

   (b) Without calculating, which player has the larger SD? Explain.
   Player B. B's points are much farther from 10.

   (c) Find $s$ for Player A.
   Differences $-1, 0, 0, 1, 0$. $\Sigma(x - \bar{x})^2 = 2$. $s = \sqrt{2 \div 4} = \sqrt{0.5} \approx 0.71$

   (d) Player B's SD is about 6.67. A coach needs a player who scores close to 10 points every game. Which player should the coach choose? Explain.
   Player A. A's SD is much smaller, so A's scores stay close to 10.

4. All three dot plots have a mean of 5. Put them in order from the **smallest** SD to the **largest** SD. Explain.




R, P, Q. The dots in R are packed close to 5. The dots in Q are far from 5.
(The SDs are about 0.67, 1.83, and 3.16.)

5. Think about what the SD means.

   (a) Can a standard deviation be negative? Explain.
   No. It comes from adding squares (never negative) and taking a square root.

   (b) Find the SD of 10, 10, 10, 10 without a formula. Explain.
   0. Every value equals the mean, so there is no spread.

   (c) Set B is 9, 10, 10, 11. Set C is 0, 10, 10, 20. Both have a mean of 10. Which set has the larger SD? Explain.
   Set C. Its values are much farther from 10.